Divya Natekar

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Education

New York University, Tandon School of Engineering — Brooklyn, NY

Master of Science — Urban Data Science, GPA: 3.61/4.0 — Sept 2024 – May 2026

- **Courses:** Data Analysis, Machine Learning, Virtual & Augmented Reality, AI, Advanced Spatial Analytics, Urban Computing

Experience

PM Accelerator — Remote

Software Engineer Intern — AI/ML, Lead Engineer — Aug 2026 – Oct 2026

- **AI Decision Engine & REST APIs:** Designed and deployed Next.js App Router API micro-routes (`/api/people`, `/api/decisions`) backed by Supabase and PostgreSQL to power a real-time AI personal relationship manager.
- **LLM Telemetry & Monitoring:** Integrated Langfuse SDK telemetry across Next.js API boundaries to trace LLM execution flows, evaluate prompt latency, and log real-time inference inputs.
- **Context Pipeline Architecture:** Engineered snapshot transformation utilities mapping asynchronous database state into structured context payloads (`User`, `Person`, `Interaction History`) for prompt evaluation.
- **Database Optimization & Security:** Refactored query logic by implementing ISO timestamp boundary filtering (`.gte`/`.lte`) in Supabase, replacing regex searches and ensuring data security across server routes.
- **Engineering Leadership & Recognition:** Awarded **Tier 1 Intern** status; led code reviews, resolved Git branch rebasing/merge conflicts, and enforced system standardization across the dev team.

L&T Technology Services (LTTS) — Airoli, India

Software Development Intern — Jan 2023 – Jul 2023

- **Backend Development:** Built consumer and backend components for a proof-of-concept healthcare data platform integrating Connected Wearable Devices with the Philips HSDP cloud ecosystem.
- **Real-Time Data Engineering:** Developed real-time data ingestion and processing pipelines using Java Spring Boot, Apache Kafka, and Spring Vault.
- **Cloud Infrastructure & DevOps:** Provisioned cloud infrastructure using Terraform/Azure and implemented CI/CD workflows in Azure DevOps with Swagger/Postman API documentation.

Projects

Agentic Retail & Ecommerce Copilot Platform — Sept 2026 – Present

Architect & Lead Developer | Python 3.13, FastAPI, LangGraph, Qdrant, PostgreSQL

- **Agentic Workflow Architecture:** Engineered an asynchronous FastAPI backend integrated with LangGraph and Google Gemini API to orchestrate tool-aware AI shopping assistants capable of executing real-time SQL product queries and inventory checks.
- **Semantic Search Pipeline:** Architected high-density vector search using Qdrant Vector DB and FastEmbed for hybrid semantic search, optimizing catalog retrieval latency for natural language product queries.
- **Async Data Engine:** Designed relational schemas using SQLAlchemy 2.0 Async ORM, asyncpg, and Pydantic v2 validation with PostgreSQL, optimizing connection pooling for high concurrency.
- **Event-Driven Checkout:** Implemented asynchronous Stripe payment processing using `PaymentIntents` endpoints with Celery, Redis task queues, and HMAC webhook verification for automated order fulfillment.

Research

Thinking on the Move: Agent-Based Urban Simulation & Decision Modeling — Sept 2025 – May 2026

NYU Center for Urban Science + Progress & Downtown Brooklyn Partnership

- **LLM Choice Modeling:** Integrated LLMs, review text embeddings, and spatial interaction engines into an agent-based simulation framework to analyze urban mobility flows and multi-agent decision behaviors.

Transient Object Removal from Urban LiDAR Point Clouds — Jun 2025 – Aug 2025

NYU CUSP Guided Research (Mentor: Prof. Debra Laefer)

- **Machine Learning Workflow:** Designed RANSAC and spatial clustering filtering workflows to remove transient urban objects from 3D LiDAR point clouds; awarded the NYU Tandon CUSP Experiential Learning Scholarship.

Technical Skills

AI Architecture & Agents: Multi-Agent Orchestration (LangGraph, LangChain), Tool Calling, RAG, Gemini API, Vector Embeddings, Langfuse, Prompt Engineering

Vector Search & Databases: Qdrant Vector DB, FastEmbed, Hybrid Search, PostgreSQL, Supabase, SQLAlchemy 2.0 Async, asyncpg, Pydantic v2

Backend & Languages: Python 3.13, FastAPI (ASGI/asyncio), REST APIs, Next.js, Java Spring Boot, TypeScript, C++, SQL

DevOps & Systems: Stripe API, Celery, Redis, Apache Kafka, Docker, Azure, Terraform, Git, GitHub, OpenAPI/Swagger, JWT

References

Debra Laefer: Professor at the NYU Tandon School of Engineering and the Center for Urban Science + Progress, New York University.